

Standard Operating Procedure

LabVIEW Camera Image Capture VI

Document type: Operating procedure

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Application: Single-image acquisition and PNG file saving from a selected camera

Document title	LabVIEW Camera Image Capture VI SOP	Version	1.0
Prepared for	Operating users / lab personnel	Prepared by	To be completed
Output format	PNG image file	Primary function	Capture and save one image on demand

1. Purpose

This SOP describes the standard method for operating the LabVIEW Camera Image Capture VI shown in the attached front panel and block diagram. The VI allows the user to select a camera, preview the acquired image, define a save location, and write the current image to disk in PNG format when the Take image button is pressed.

2. Scope

This procedure is intended for laboratory users, operators, and students who need to capture a still image from the configured camera system and save it to a specified folder. It applies to routine single-image capture and basic verification of successful image saving.

3. Pre-startup checks

- Confirm that the required camera is physically connected and visible in NI MAX or the Vision acquisition system.
- Confirm that the intended save folder already exists and that the user has write permission to that folder.
- Check that the file name entered in File Path ends with the desired image extension, for example .png.
- Confirm that no other application is exclusively locking the camera.

4. Front panel controls and indicators

Item	Function	Operator guidance
Camera selection	Selects the camera source used for acquisition.	Choose the correct camera before running the VI; verify it matches the hardware intended for use.
File Path	Defines the folder location and output file name for the saved image.	Use a valid folder path and a unique file name to avoid accidental overwrite.
Take image	Triggers writing of the current image to disk.	Press after the image preview is acceptable and the File Path has been confirmed. Double click on the green button to capture and save the image.
Image Quality	Passes a quality/encoding parameter to the image write function.	Keep the validated default unless your application requires a different setting.
Image display	Shows the currently acquired image from the selected camera.	Use this window to confirm focus, field of view, and general image acceptability before saving.

5. Operating procedure

- 1. Open the VI:** Launch the LabVIEW VI and verify that the front panel loads without errors.
- 2. Select the camera:** From the Camera selection control, choose the required camera, for example cam6.
- 3. Enter the output path:** In File Path, enter the complete destination including file name, for example C:\Digambar\SMASYS\Camera\test_1.png.
- 4. Set the quality value:** Review the Image Quality control and keep the approved default value unless a different value has been specified for your setup. Keep it at 1000 so there is no image compression happening.
- 5. Run the VI:** Click the Run continuously arrow in LabVIEW. The image window should populate with the current view from the selected camera.
- 6. Verify the preview:** Inspect the image display feed and confirm that the correct part, field of view, and exposure are visible.
- 7. Capture the image:** Press the Take image button twice to capture and save the current image.
- 8. Verify file creation:** Open the destination folder in Windows Explorer and confirm that the PNG file was created successfully.
- 9. Repeat if required:** If additional images are needed, update the file name before pressing Take image again so earlier images are not overwritten.
- 10. Stop and close:** After the required capture is complete, stop the VI and close the application in the normal way.

Important: If the same file name is reused, the previous image may be overwritten. Use a unique file name for each capture unless deliberate replacement is intended.

6. Troubleshooting

Issue	Likely cause	Recommended action
No image appears in the display	Camera is not connected, not recognized, or the wrong camera was selected.	Check hardware connection, verify the camera in NI MAX, and reselect the correct camera.
Image does not save	Invalid File Path, missing folder, or no write permission.	Create the folder first, verify the path syntax, and save to a writable location.
Saved file cannot be found	File name or folder path was entered incorrectly.	Copy the path directly from Windows Explorer or browse to the folder before capture.
Wrong camera feed is shown	Incorrect camera was chosen in Camera selection.	Stop the VI, choose the correct camera, and run the VI again.
Existing image was replaced	The same output file name was reused.	Use unique names such as layer_1, layer_2, or add a date-time suffix.

7. Functional summary of the VI

- The block diagram initializes the selected camera session and creates an NI Vision image buffer.
- The VI runs inside a loop so the operator can interact with the front panel while the image is displayed.
- A case structure checks the Take image command. When true, the current image is written to the path entered by the user.
- The image write node is configured for PNG output in the current implementation.
- Error wires are used to propagate acquisition and file-writing errors through the program.

Appendix A. Reference screenshots

A1. Front panel

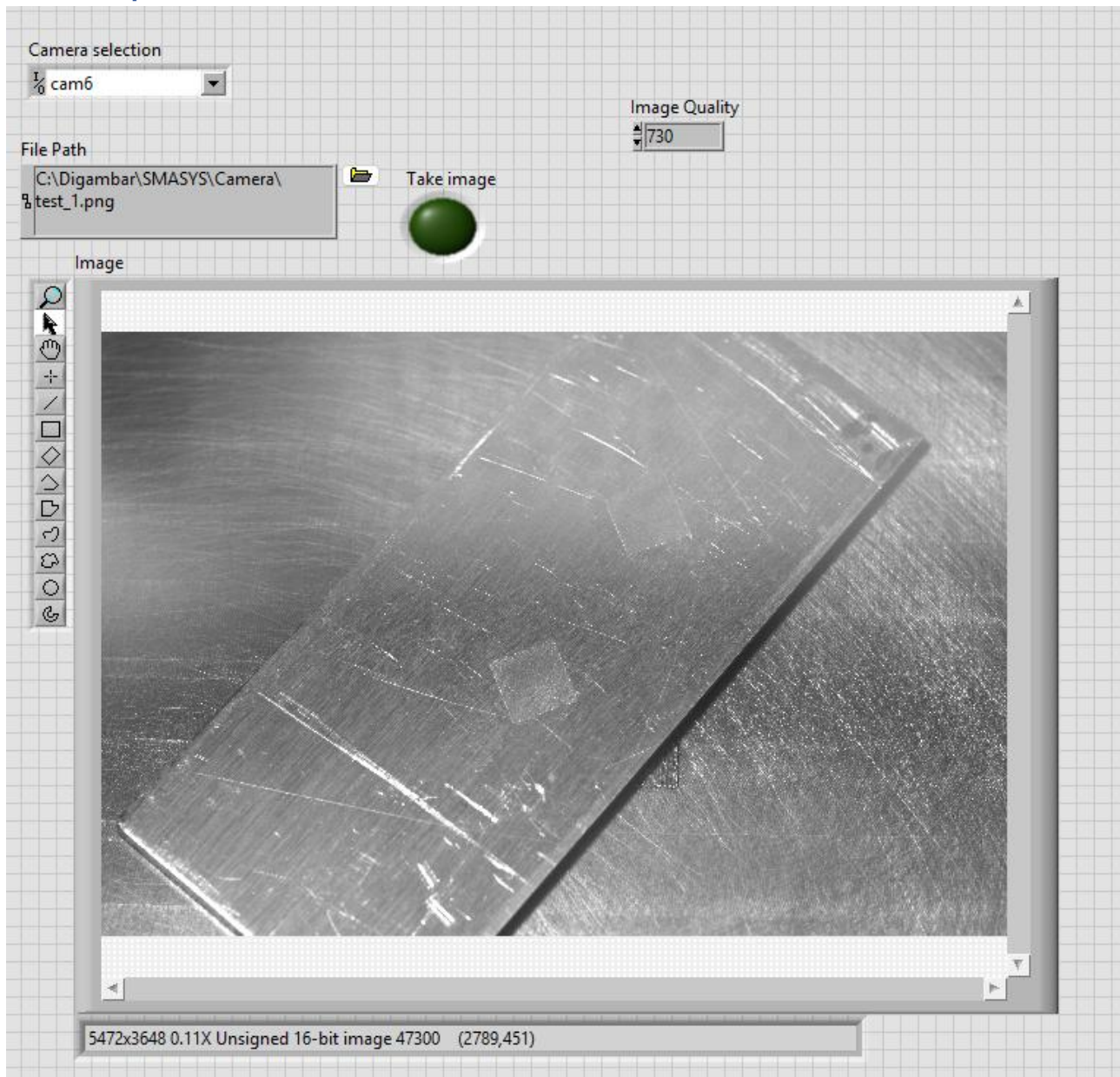


Figure A1. Front panel used for camera selection, image preview, and manual image capture.

A2. Block diagram

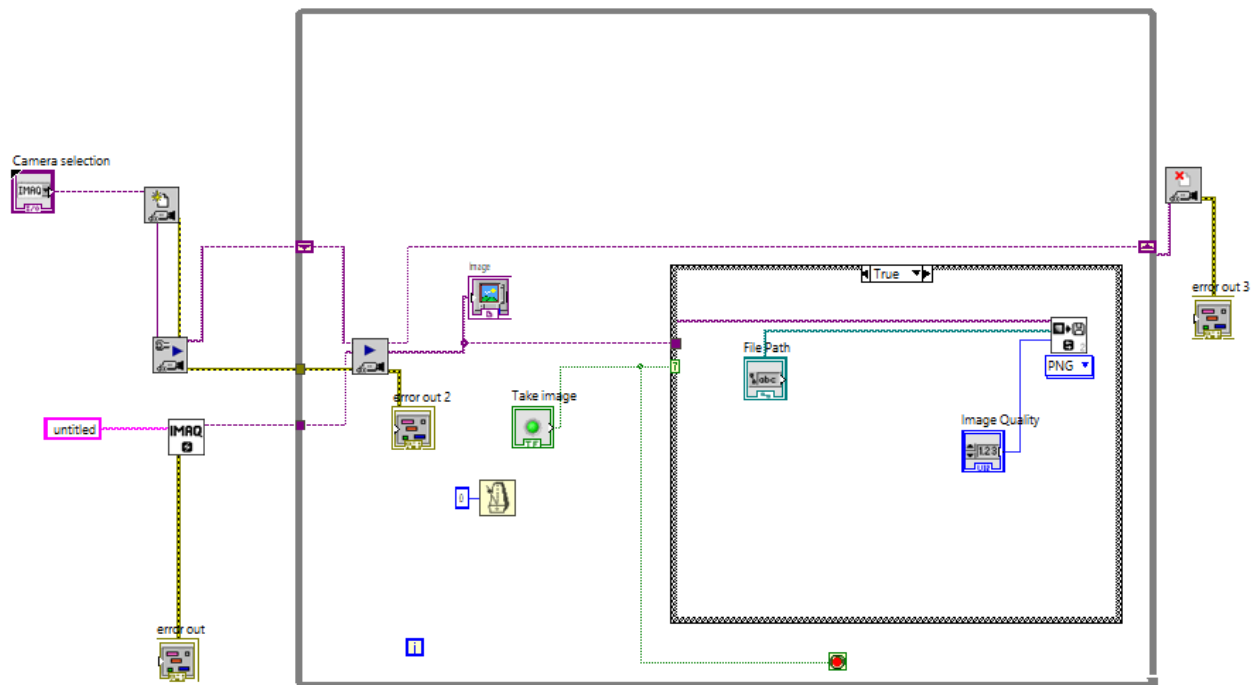


Figure A2. Block diagram showing camera initialization, image display, Take image logic, and PNG file writing.